

Package #161

Incorporate a needle-free, hermetically sealed port with a self-sealing membrane at the catheter hub, creating a closed fluid circuit that eliminates open lumens during dialysis connections.

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'flow patency loss' concerning tunneled hemodialysis catheter. User-submitted engineering problem (session input): "Why do tunneled hemodialysis catheters lose flow patency within weeks of insertion despite flushing protocols?". [fda_maude] 100 records retrieved live (1992-2032). Illustrative records: 'MAUDE report 1000: Injury [SAME AS ABOVE]'; 'MAUDE report 10000024: Injury [CAPSURE SP NOVUS LEAD]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2015-2026). Illustrative records: 'Heparin-Coated Tunneled Hemodialysis Catheters Improve Failure-Free Survival in ESRD.'; 'The Temporal pathological changes of venous catheter-related fibrin sheath and their implications for nursing practice.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Closed-system technologies that isolate the fluid pathway to reduce contamination risk.

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.